Mosquito Lagoon Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

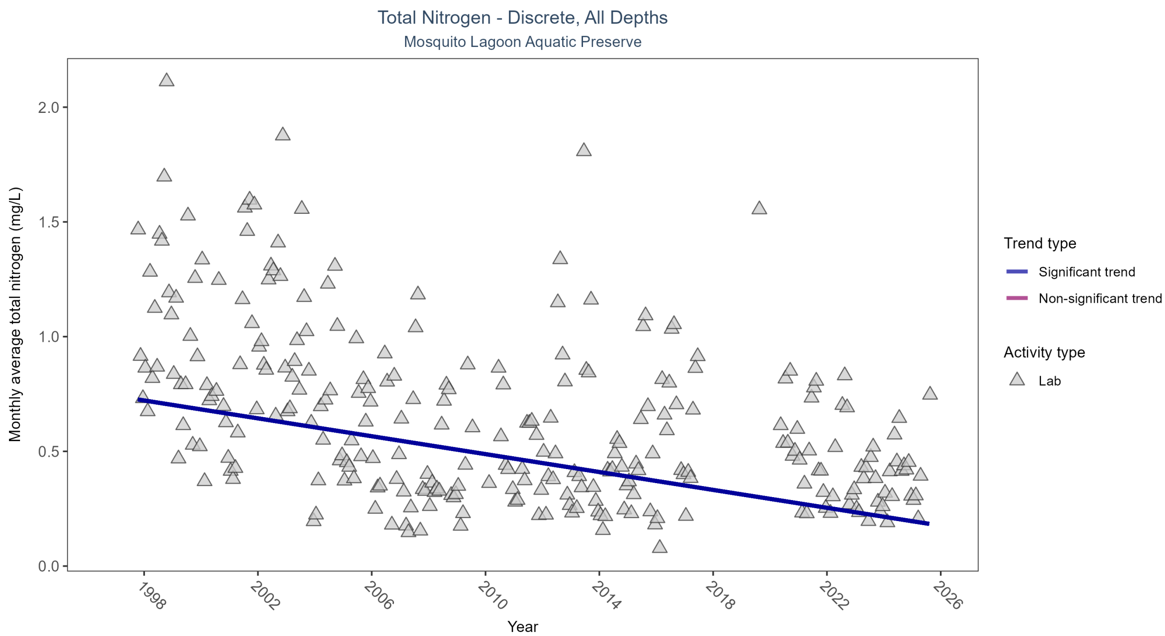

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen decreased by 0.02 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	386	28	1997 - 2025	0.68912	-0.42119	0.74129	-0.01949	0

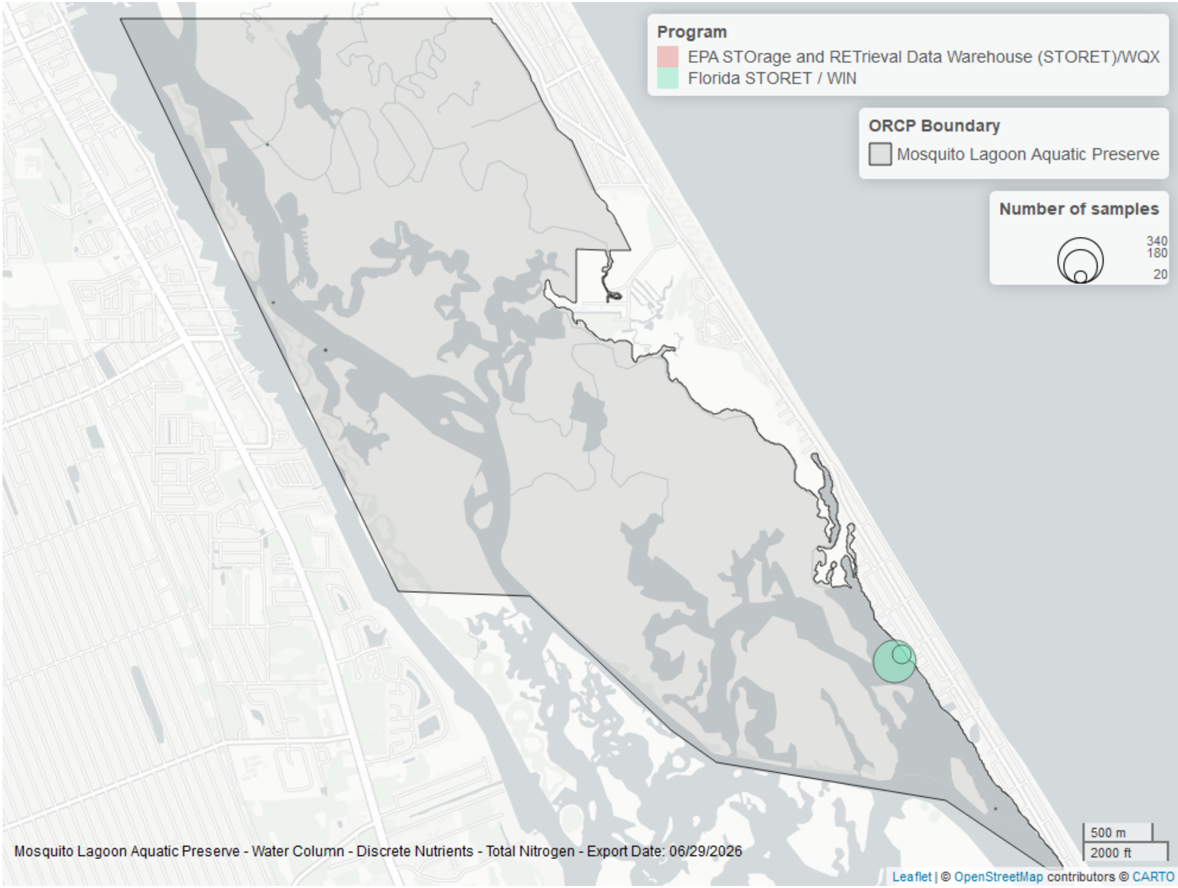

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

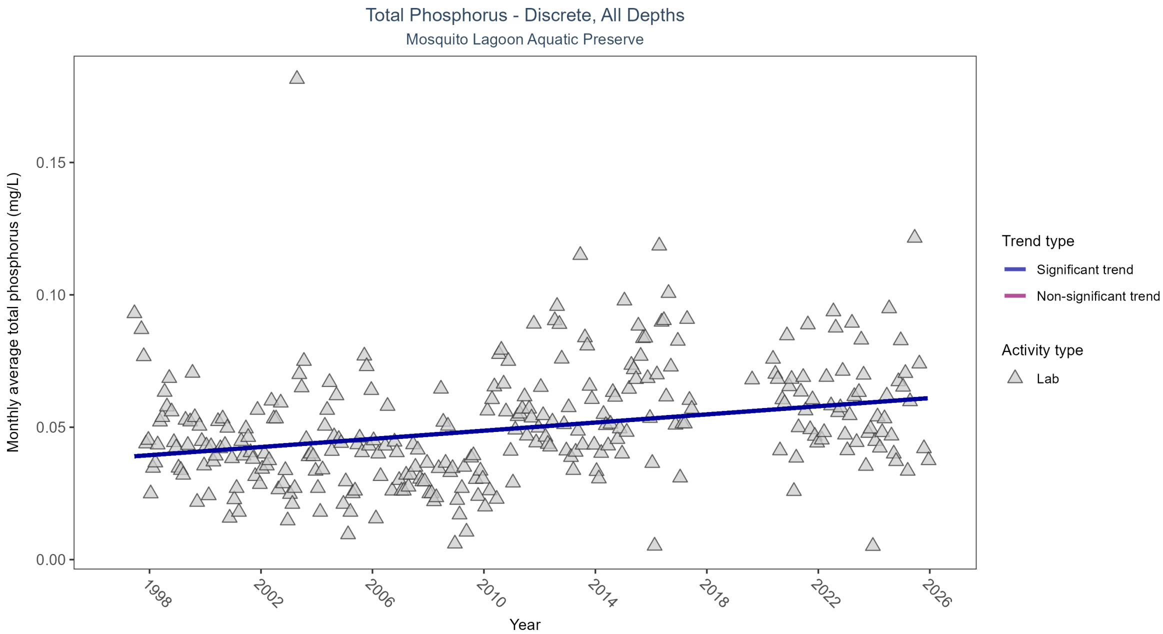

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	862	28	1997 - 2025	0.0422	0.25298	0.03868	0.00077	0

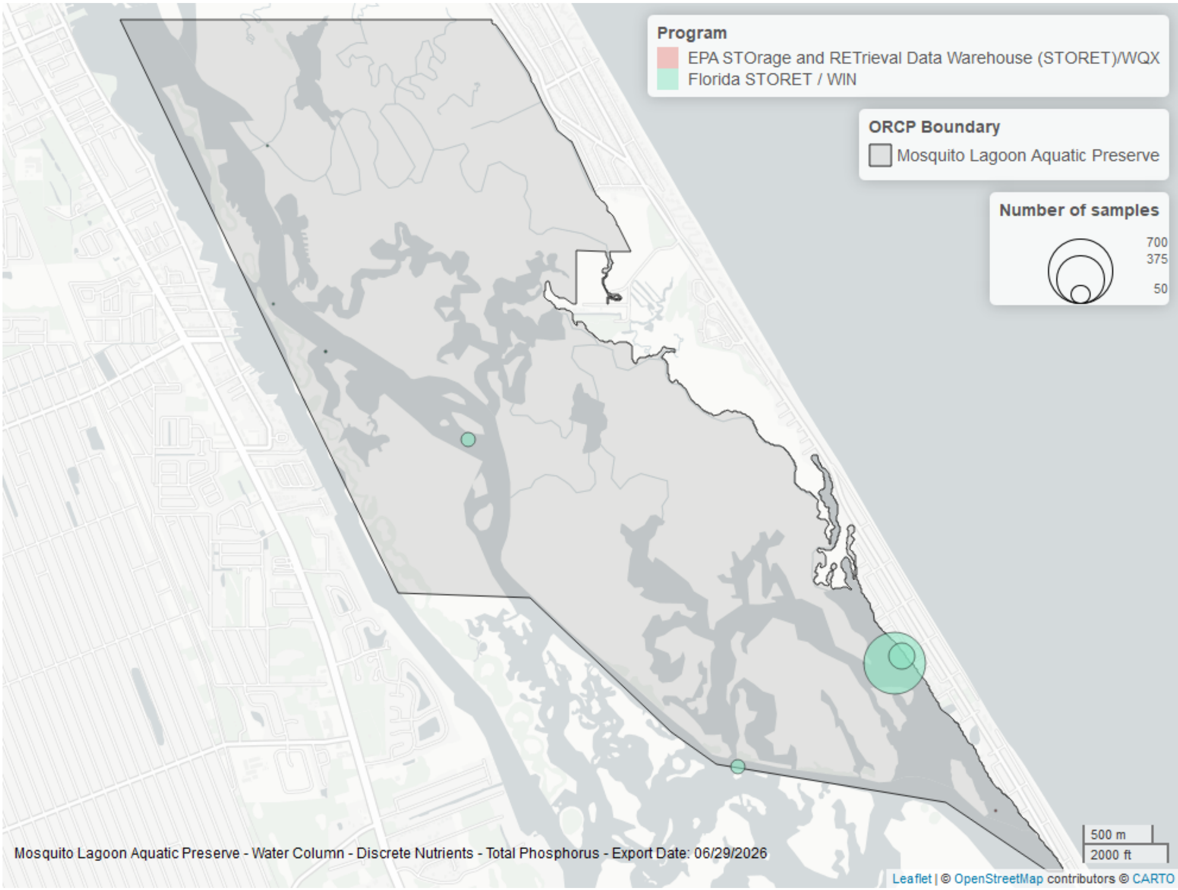

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

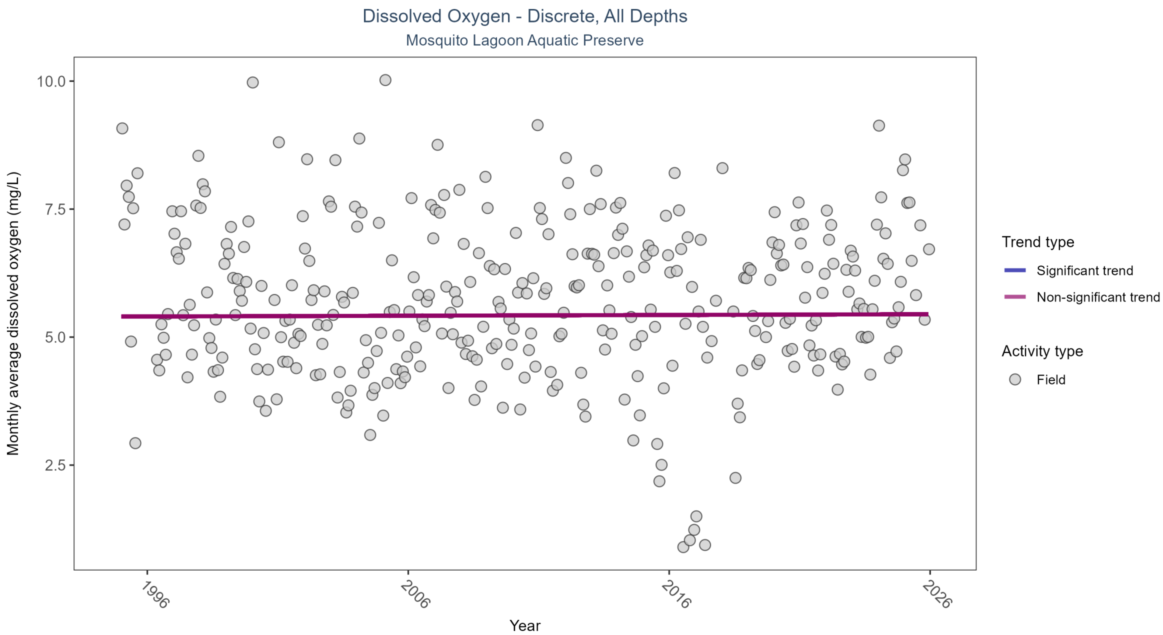

Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Dissolved oxygen showed no detectable trend between 1995 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	4810	31	1995 - 2025	5.4	0.00785	5.40124	0.00147	0.853

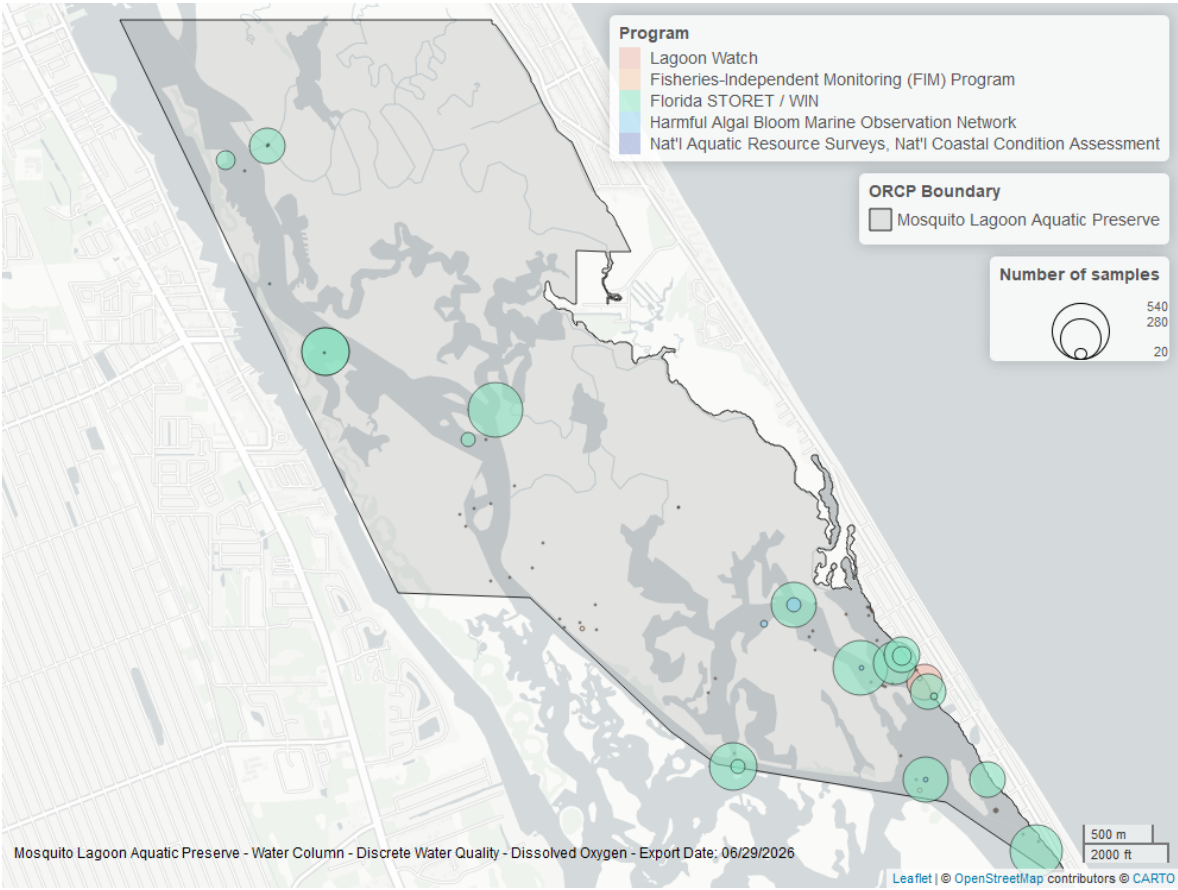

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

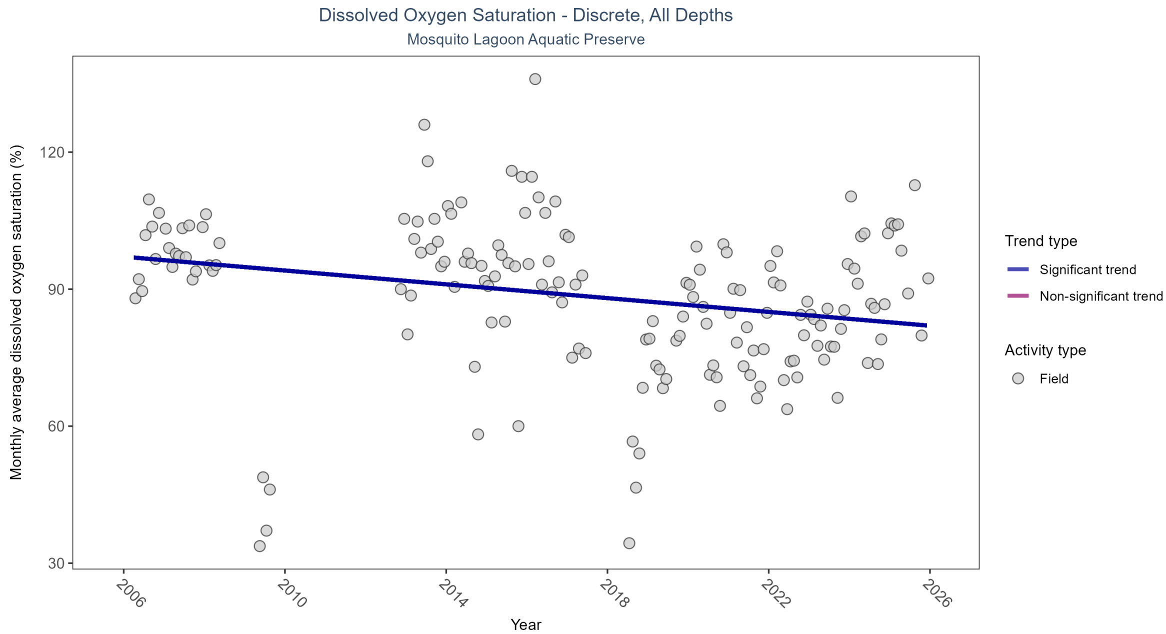

Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation decreased by 0.76% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	505	18	2006 - 2025	86.7	-0.17738	97.10367	-0.75556	0.0028

\end{table}

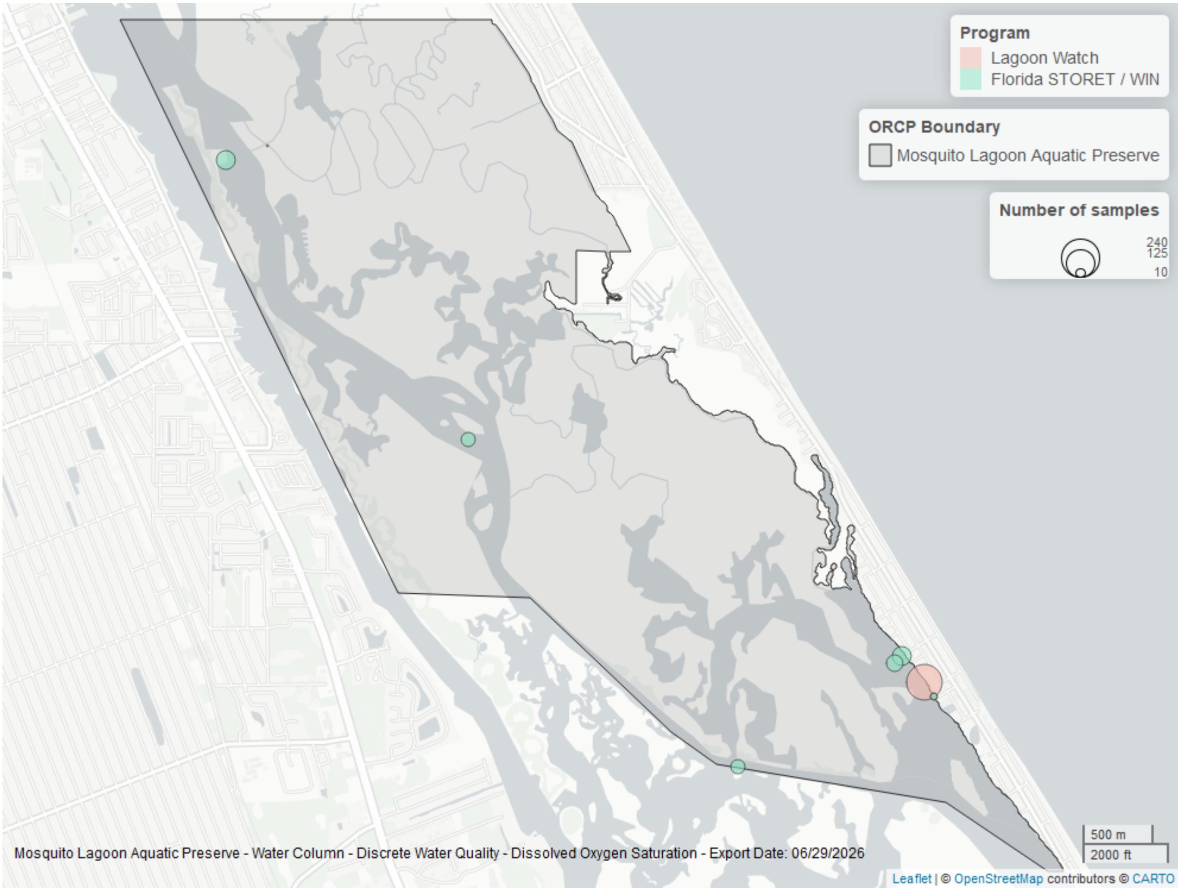

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

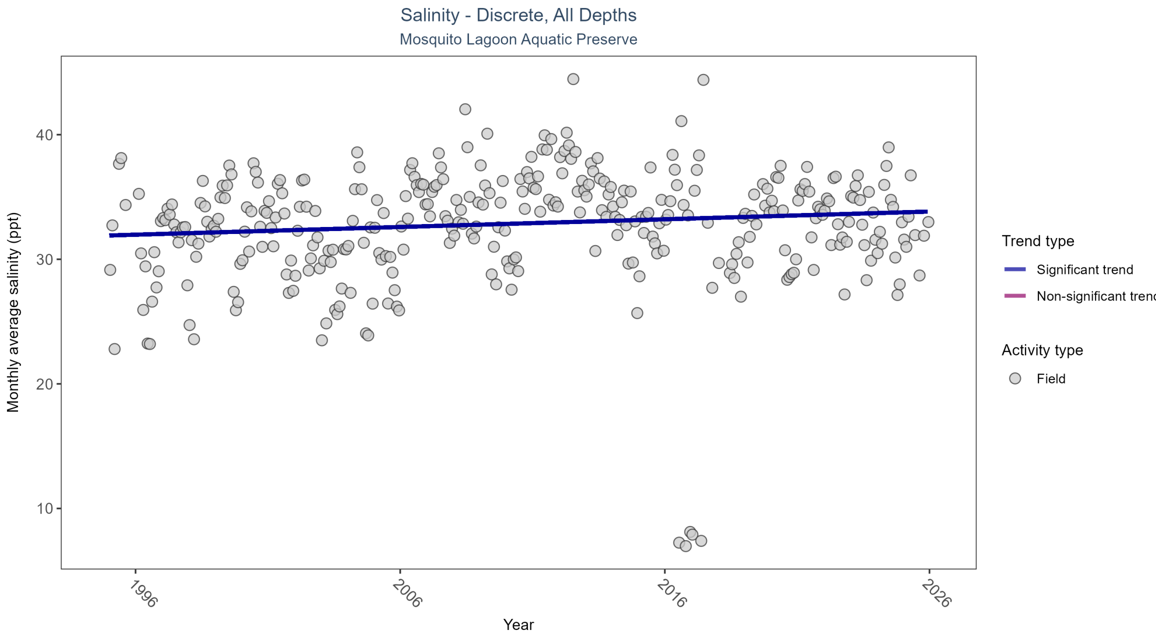

Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 4: Monthly average salinity increased by 0.06 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	5269	31	1995 - 2025	32.5	0.11041	31.91091	0.06188	0.0036

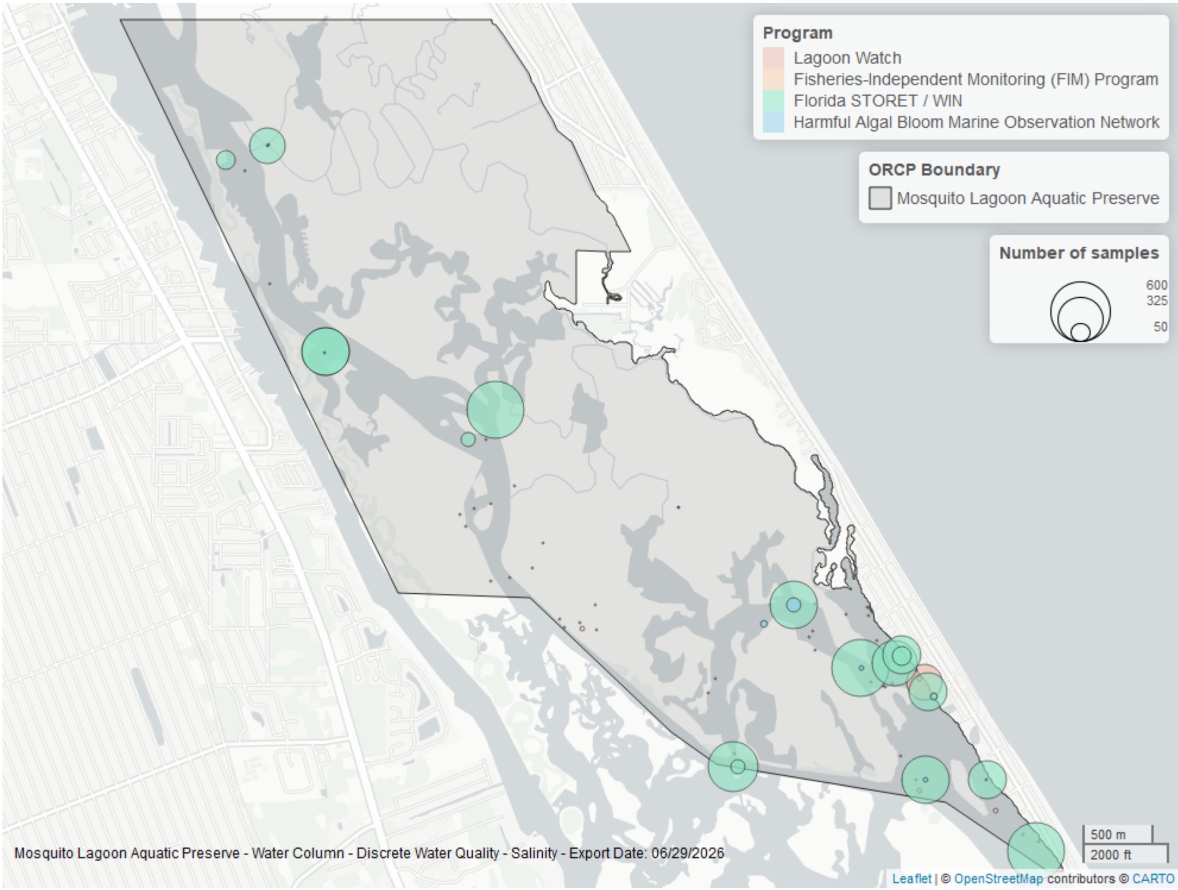

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

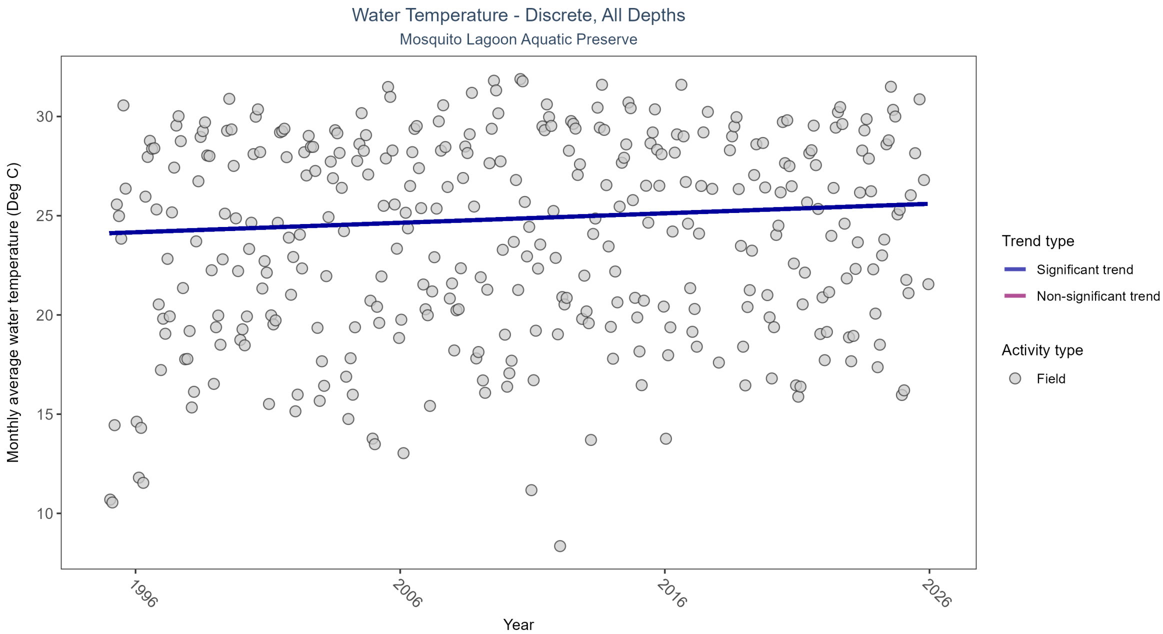

Figure 11: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 5: Monthly average water temperature increased by 0.05Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	5359	31	1995 - 2025	25.04	0.1657	24.11874	0.0478	0

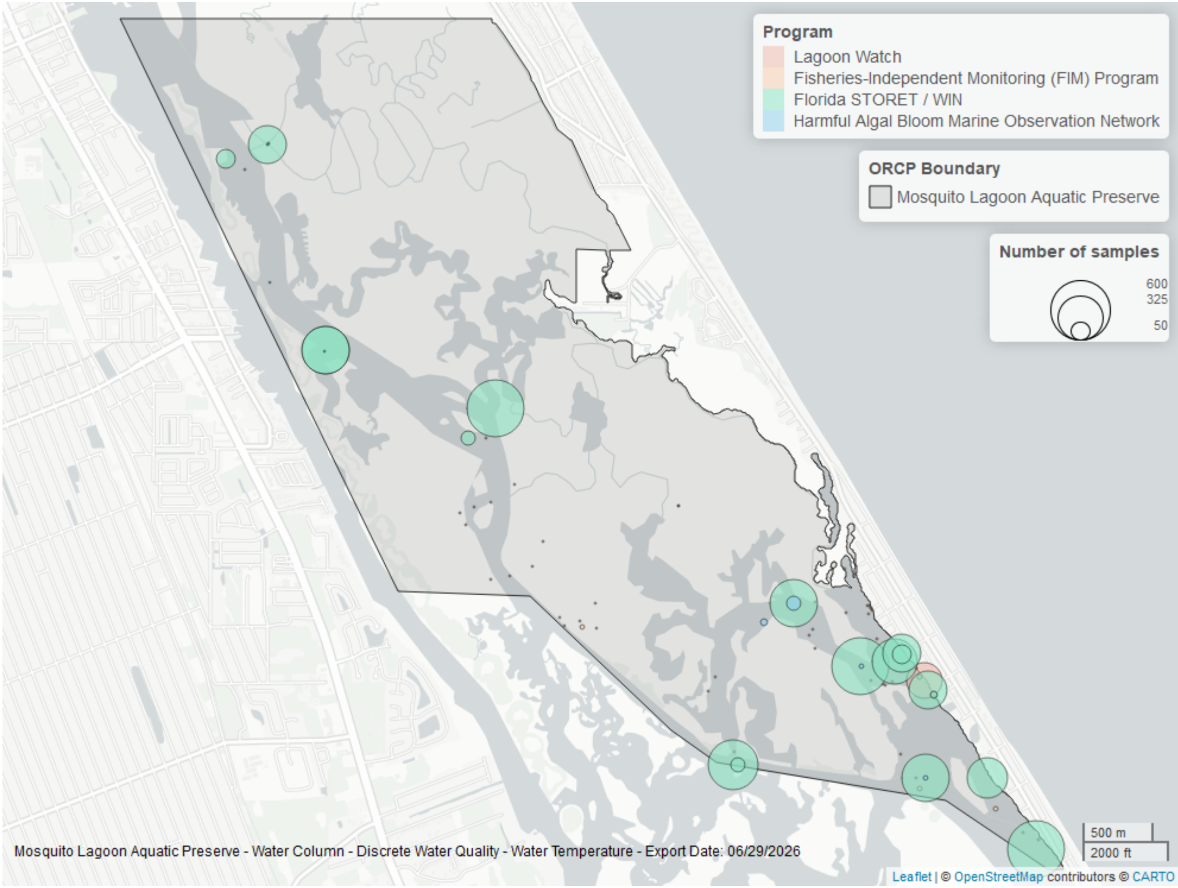

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

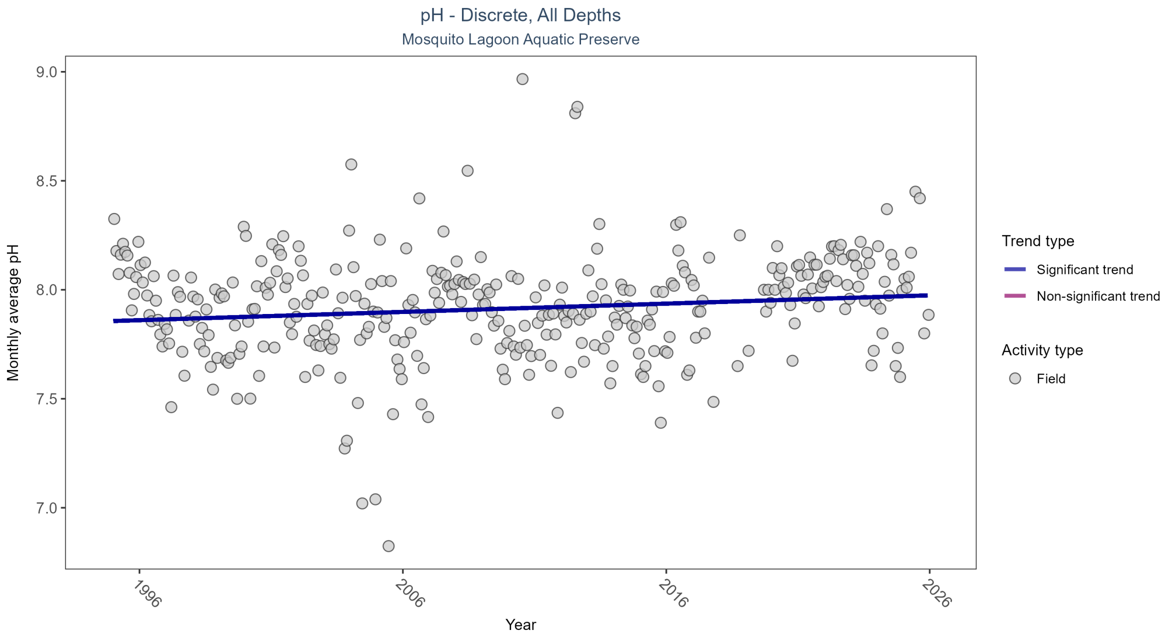

Figure 13: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 6: Monthly average pH increased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	3725	31	1995 - 2025	7.9	0.11451	7.8567	0.0038	0.0028

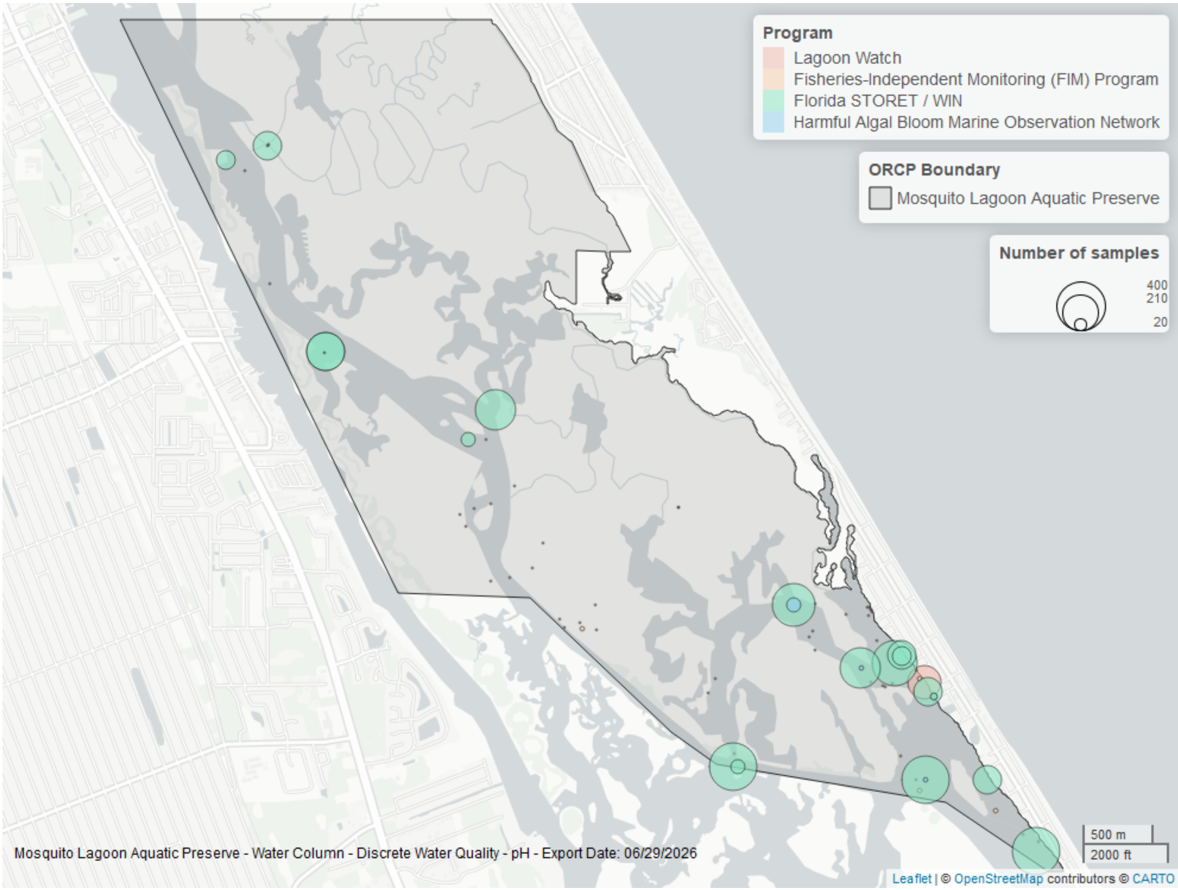

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

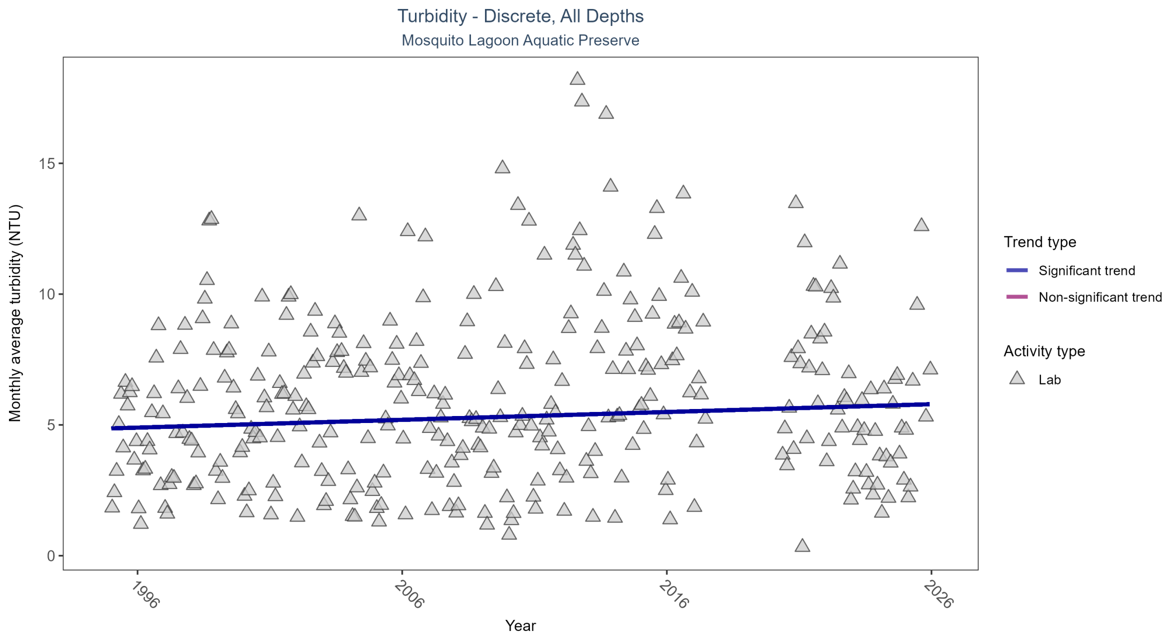

Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 7: Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3519	29	1995 - 2025	4.6	0.0901	4.86621	0.02986	0.0198

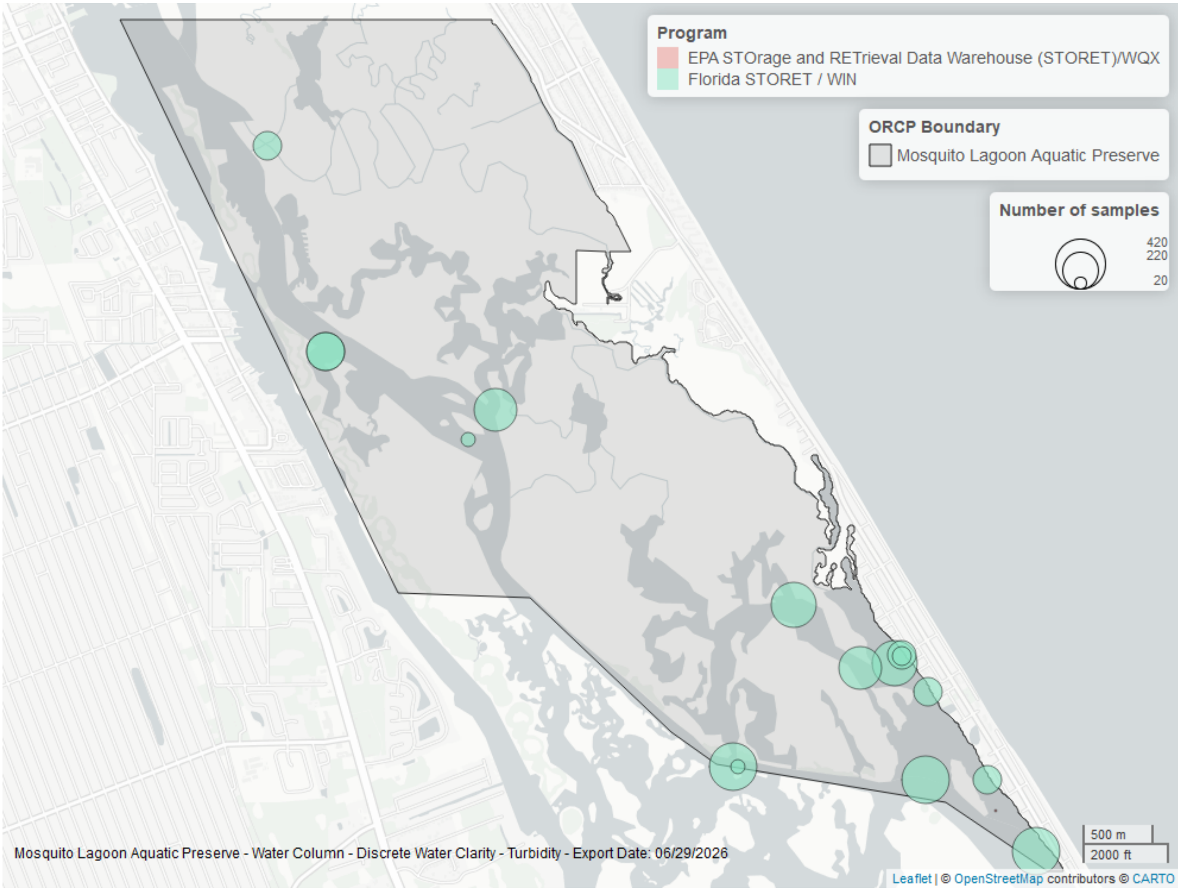

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

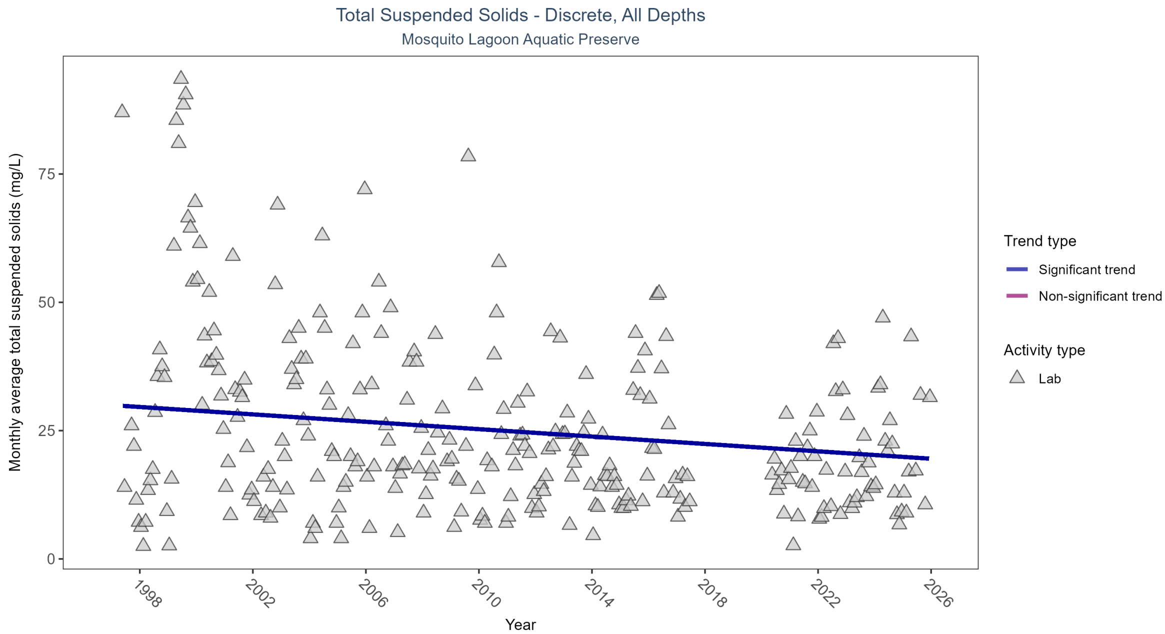

Figure 17: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 8: Monthly average total suspended solids decreased by 0.36 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	452	27	1997 - 2025	19.4	-0.17258	29.96786	-0.36	0.0001

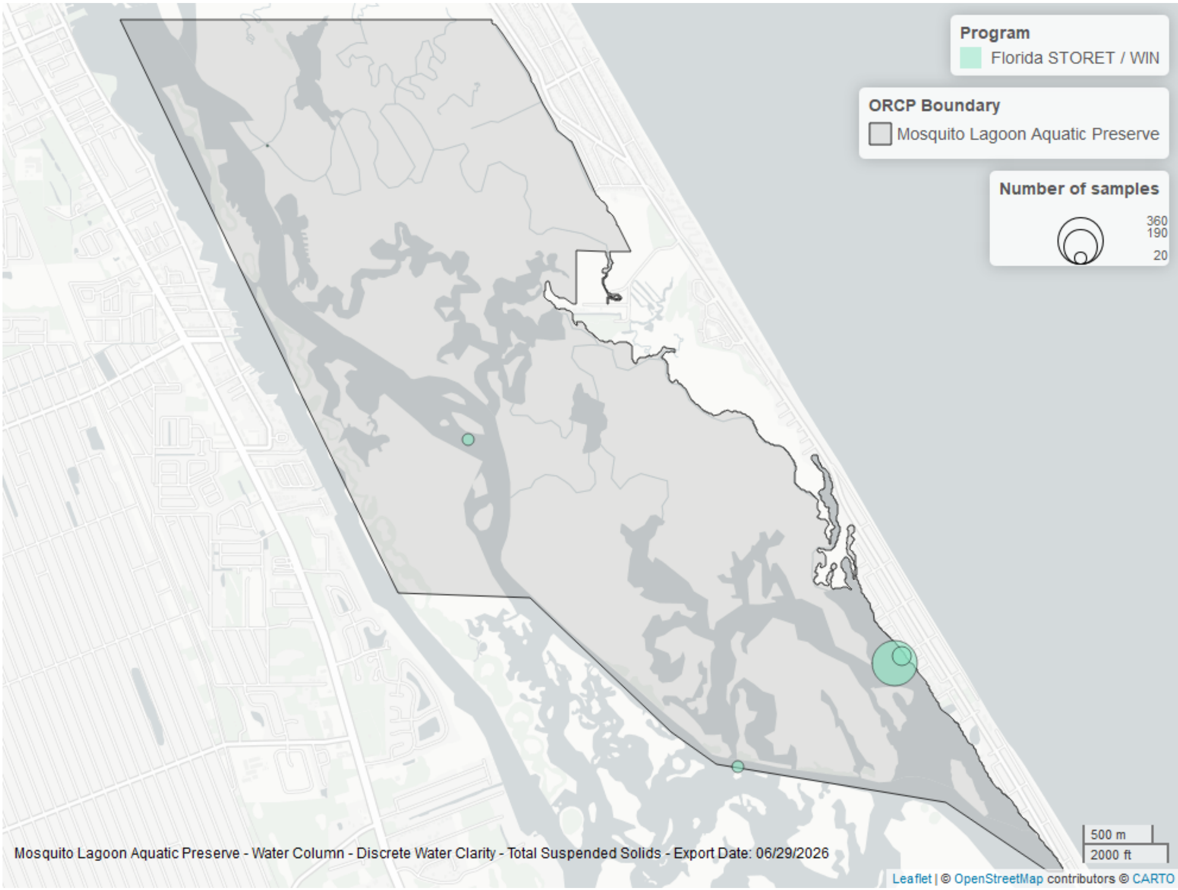

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

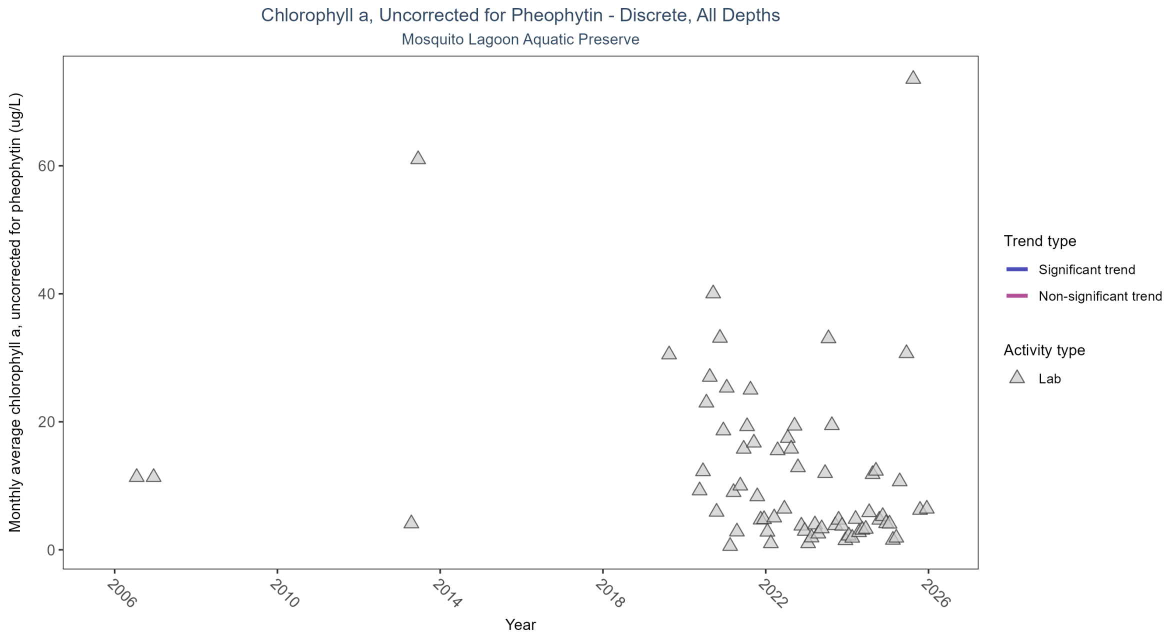

Figure 19: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 9: There was insufficient data to fit a model for chlorophyll a, uncorrected for pheophytin.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	130	9	2006 - 2025	5.93689	-	-	-	-

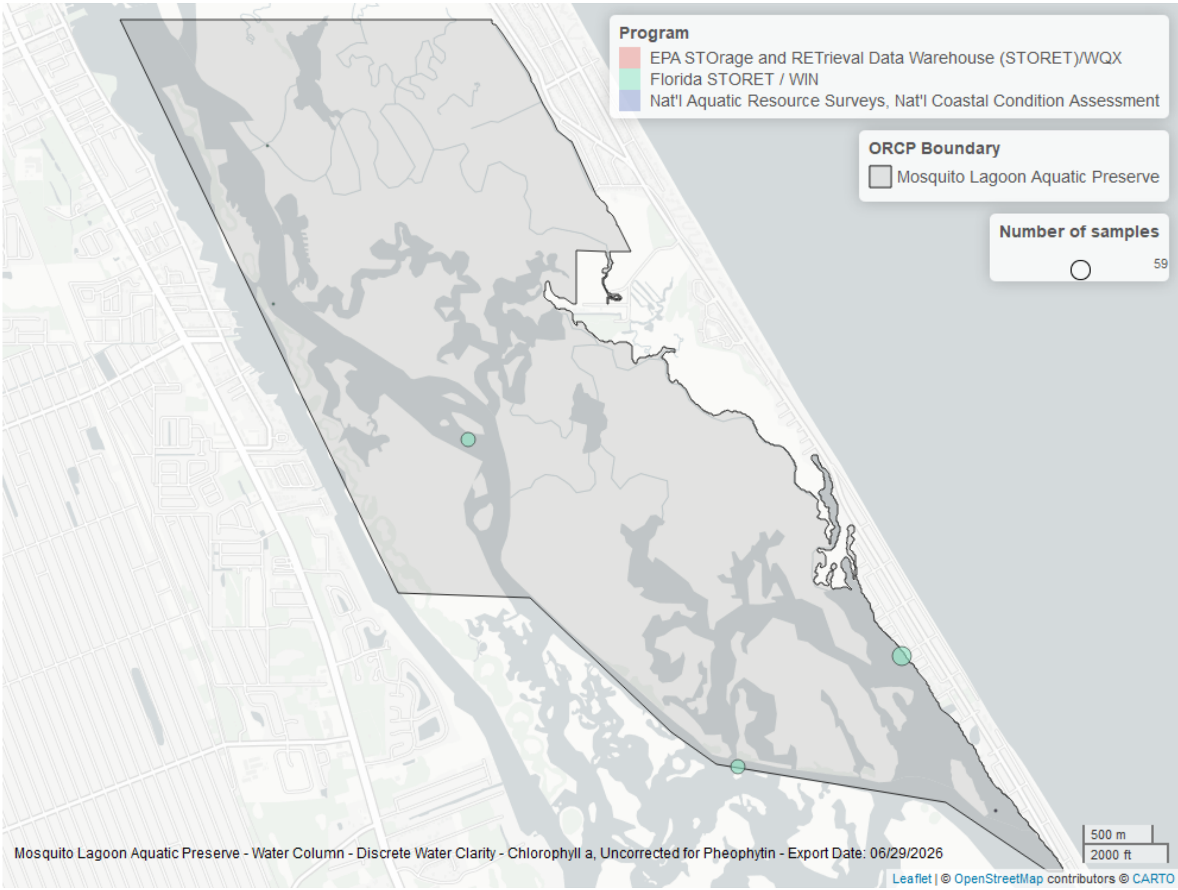

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

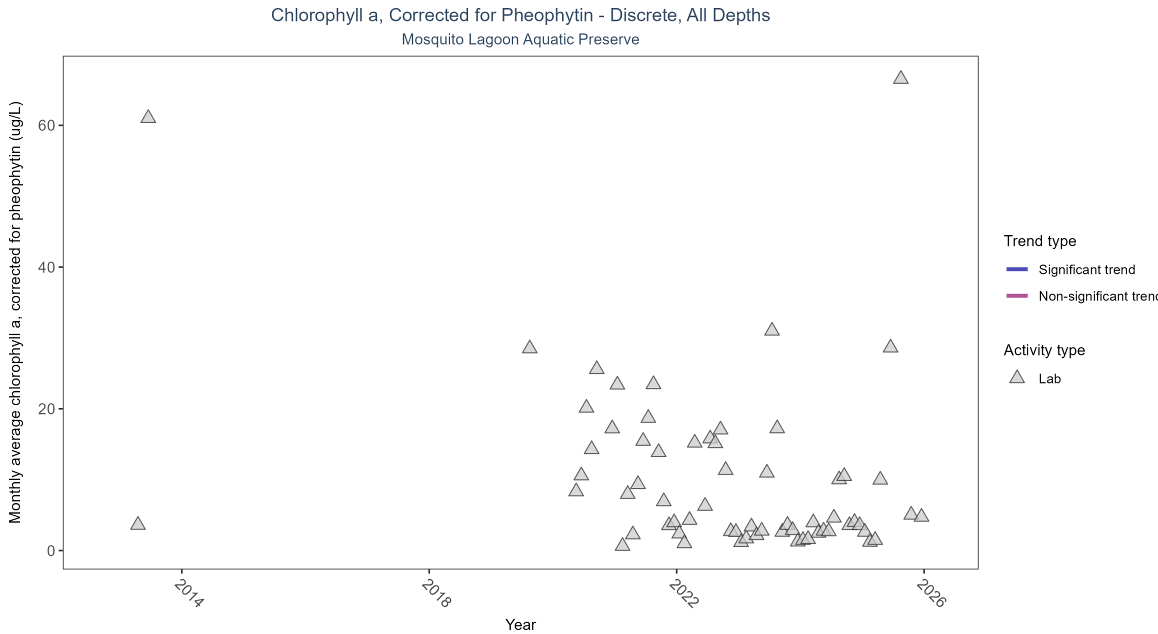

Figure 21: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 10: There was insufficient data to fit a model for chlorophyll a, corrected for pheophytin.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	125	8	2013 - 2025	4.77263	-	-	-	-

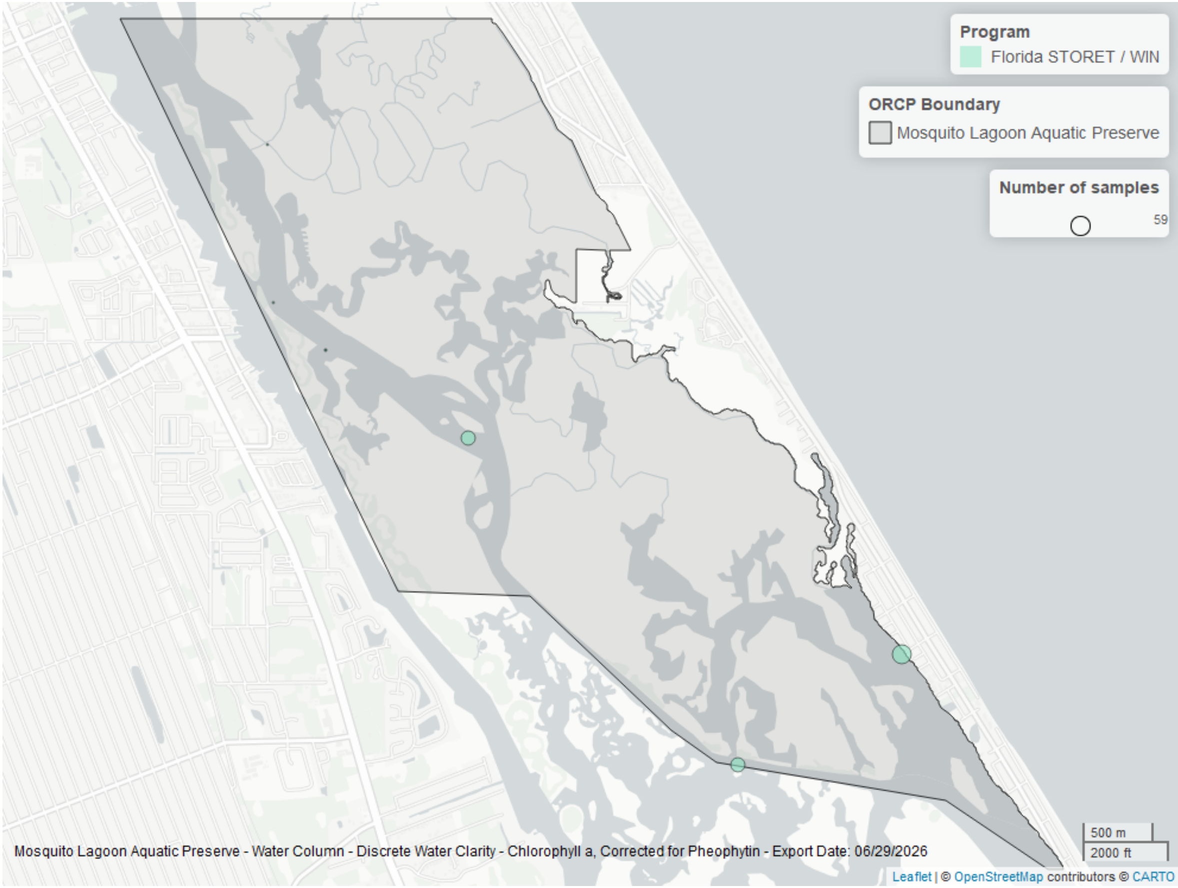

Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

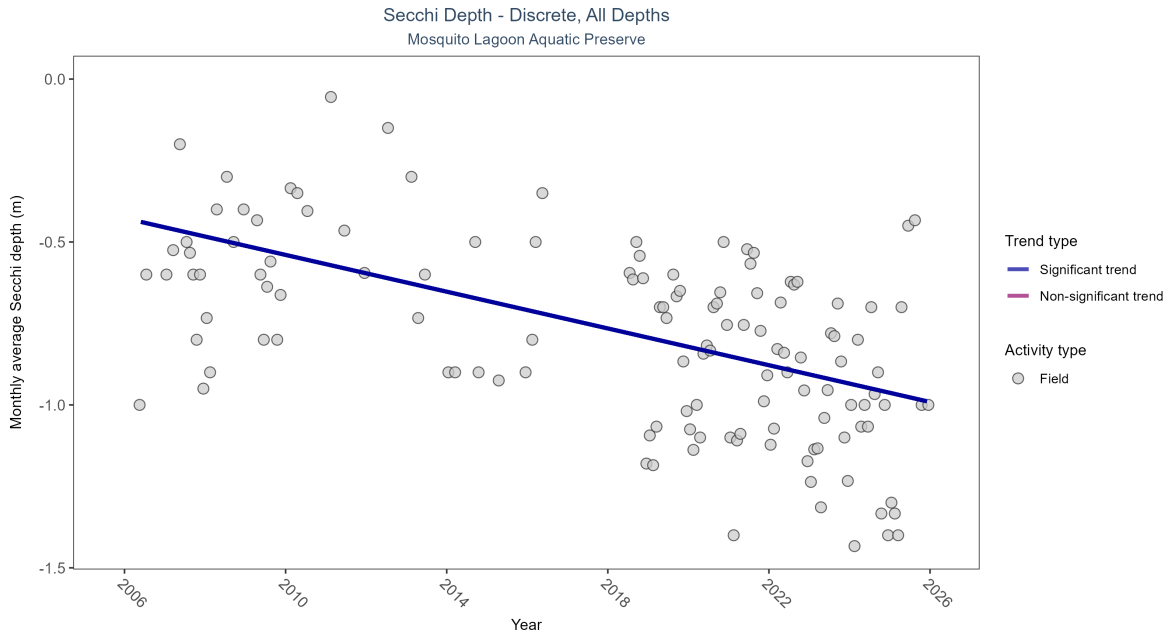

Figure 23: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 11: Monthly average Secchi depth became deeper by 0.03 m per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	694	19	2006 - 2025	-0.8	-0.46541	-0.42681	-0.02821	0

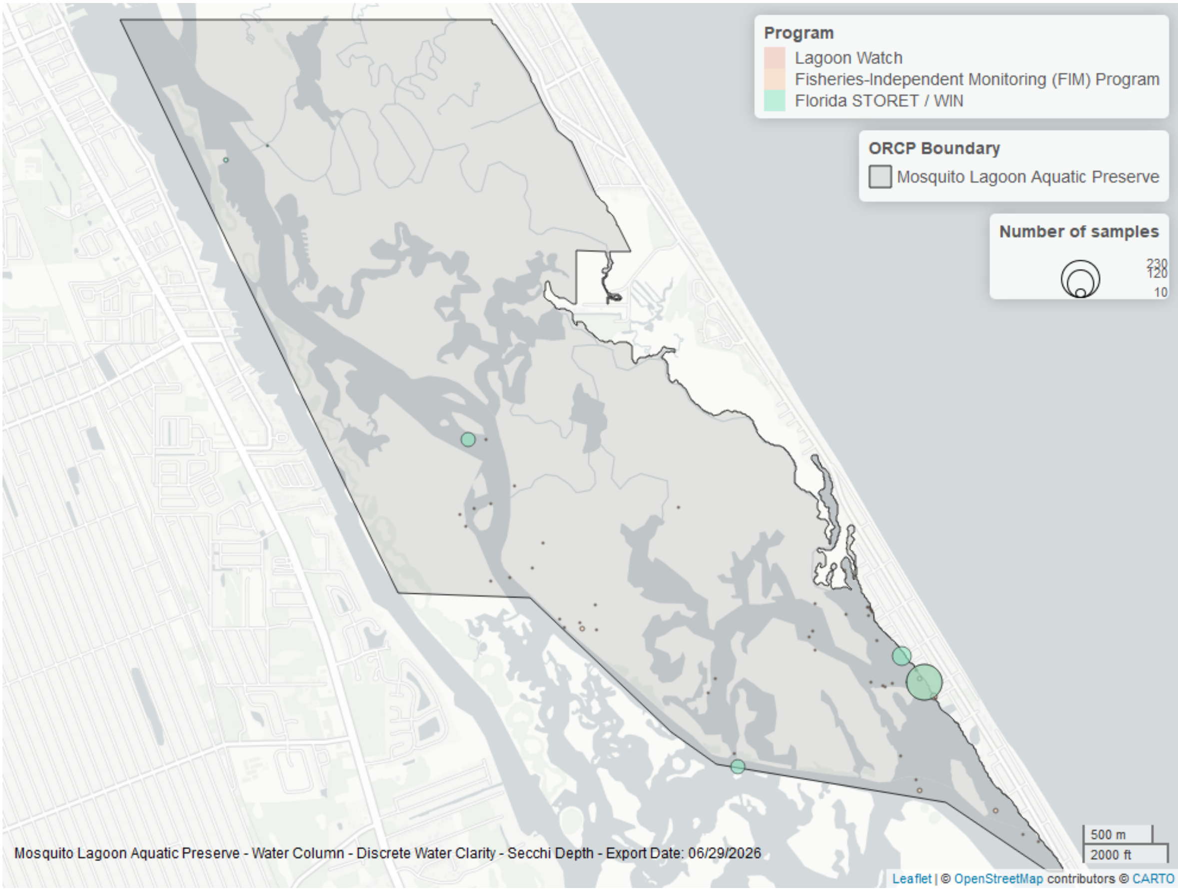

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

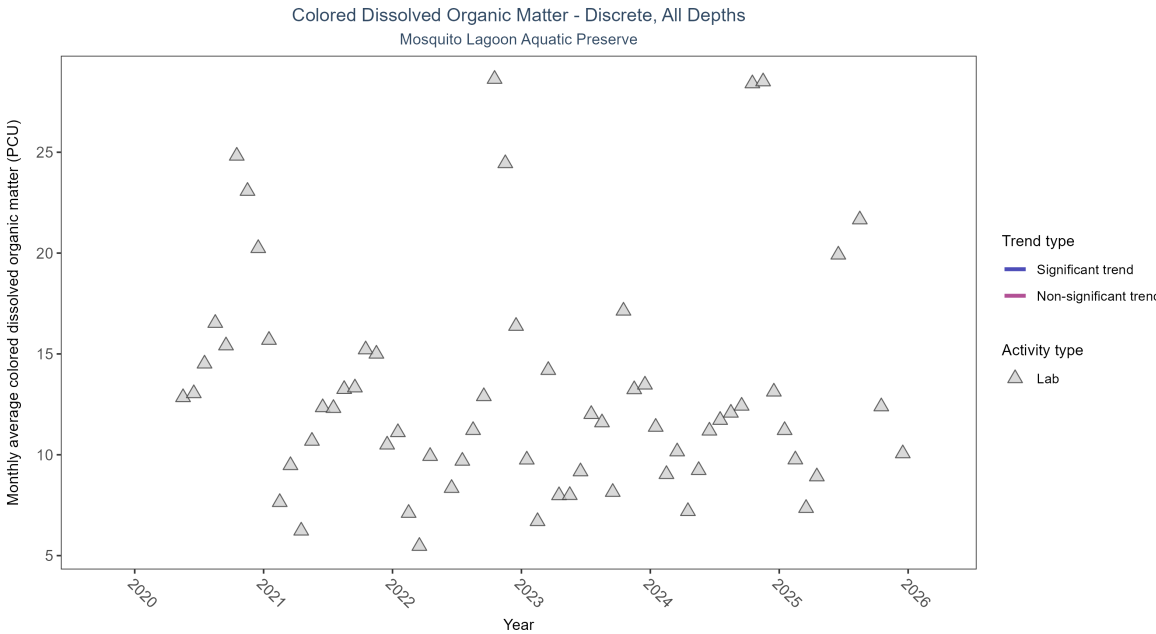

Figure 25: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 12: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	127	6	2020 - 2025	11.73611	-	-	-	-

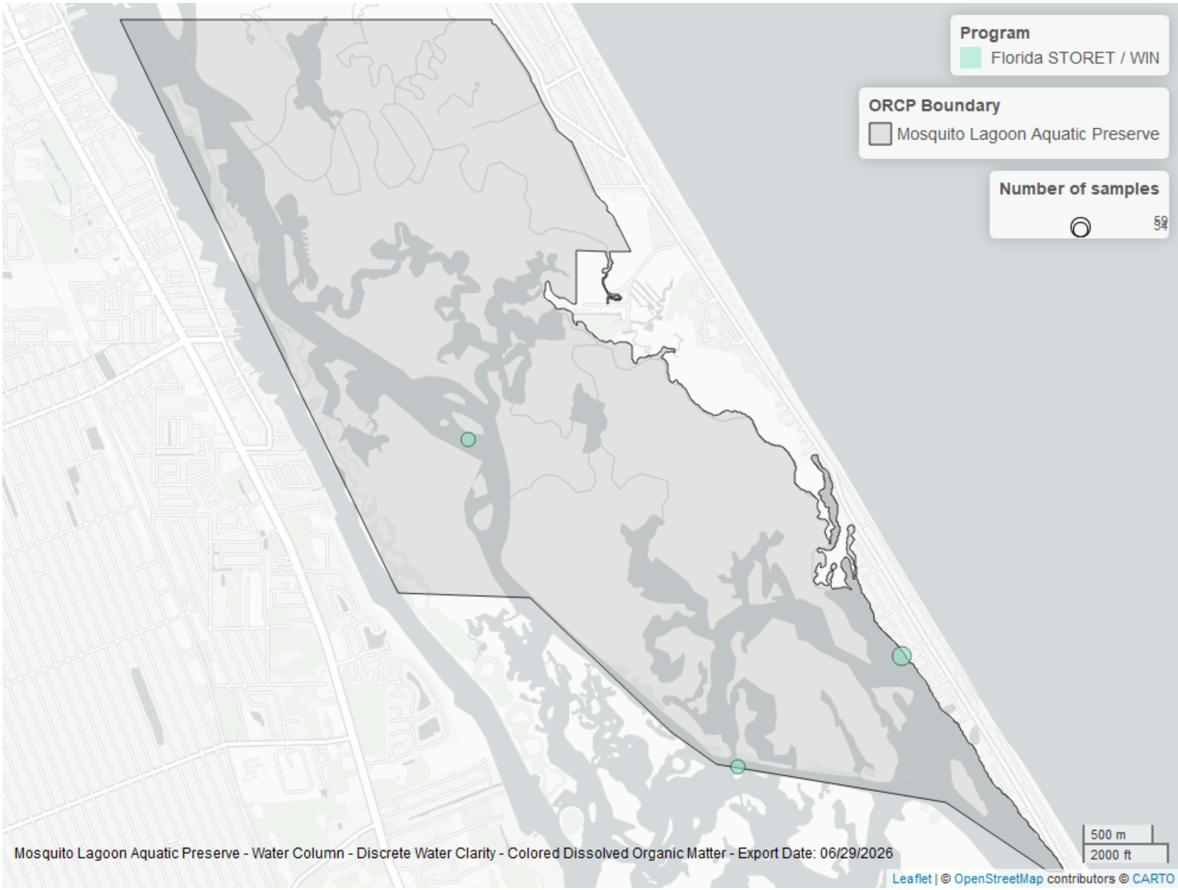

Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.